

NAGARGOJE UTTAM

Hyderabad, Telangana, India | nagargojeuttam143@gmail.com | +91 93466 22782
LinkedIn: linkedin.com/in/nagargoje-uttam-95b710302 | GitHub: github.com/NAGARGOJE-UTTAM

PROFESSIONAL SUMMARY

Early-career Software Engineer and AI-focused Computer Science student building practical software systems across backend development, machine learning, computer vision, Generative AI and Agentic AI. Hands-on work includes a predictive waste-collection platform, a from-scratch Mask R-CNN implementation, and a Java-based Telegram assistant. Strong interest in backend engineering, AI engineering, system design, and solving real-world problems through software.

TECHNICAL SKILLS

Programming: Java, Python, SQL, C++

Software Engineering: Backend Development, REST APIs, API Integration, Automation, Git, GitHub

AI / ML: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Generative AI, Agentic AI

Frameworks / Libraries: TensorFlow, Keras, OpenCV, NumPy, Pandas, Scikit-learn

Web / Database: HTML, CSS, JavaScript, MySQL

Tools: VS Code, Maven, Command Line

PROJECTS

WasteWiseAI — Predictive & Carbon-Aware Waste Collection Platform

IARE Hackverse Series — NEXORA 2026 | Final Round | September 2026

- Built a hackathon prototype for demand-driven municipal waste collection instead of purely fixed schedules.
- Designed the intelligence flow: SENSE → PREDICT → PRIORITIZE → OPTIMIZE → EXPLAIN → DISPATCH → MEASURE.
- Added explainable recommendations and scenarios for normal days, festival/surge conditions, heavy rain, traffic congestion and unit failure.
- Live Demo: wastewise-ai-null.vercel.app

Mask R-CNN From Scratch

Computer Vision / Deep Learning | August 2026 – Present

- Implementing Mask R-CNN step by step to understand the architecture rather than relying on an existing implementation.
- Built a COCO dataset viewer and worked with COCO annotations, object categories, bounding boxes, polygons, RLE and instance-segmentation masks.
- Developing the architecture around ResNet-50, Feature Pyramid Network (FPN), Region Proposal Network (RPN), ROI Align, detection head and mask head.
- Using Python, TensorFlow and Keras to study tensor flow, model components and deep-learning architecture through implementation.

IARE Assistant Bot

Java Backend / Telegram Bot

- Built a Telegram-based virtual assistant for IARE students to provide course, department, fee, hostel and admission information.
- Implemented the bot using Java, Maven and Telegram Bot API with backend development, API integration and automation.

Integrated Information Platform for Indian Universities

Information Platform / Database Application

- Designed a centralized platform concept for university information including admission procedures, ECET/MCET ranks, cut-offs and academic details.
 - Focused on organizing fragmented university information into a structured and searchable system.
 - Used database-oriented application design to manage institutional information.
-

EDUCATION

Bachelor of Technology (B.Tech) — CSE — Institute of Aeronautical Engineering (IARE), Hyderabad Expected 2028 | CGPA: 7.56/10

Diploma — Computer Science and Engineering — AITS || Aug 2022 – Jun 2025 | Grade: 8.89/10

School Education — Mhatama Jyotiba Phule Telangana BC Welfare || Jul 2013 – Dec 2022 | Grade: 8.5/10

CERTIFICATIONS

• **Oracle Certified Foundations Associate — Agentic AI** — Oracle | Issued Aug 2026 | Valid until Aug 2028

• **GenAI Powered Data Analytics Job Simulation — Certificate of Completion** — Forage | Issued May 25, 2026 | Enrolment Verification Code: 5jgwgNXihrJEwMkfh | User Verification Code: 6a14625c2264d02066aee7f3

ACHIEVEMENTS

- Final Round — IARE Hackverse Series: NEXORA 2026, a 48-hour hackathon focused on innovation and real-world problem solving.
- Published the first working stage of Mask R-CNN From Scratch on GitHub.